Transistor Family Primer — What Changed, and Why CPUs Use CMOS

This is a curiosity appendix, not a required electronics detour. Its purpose is to place the MOSFET used in the CPU lessons among the other transistor families you may remember.

First, what job are we asking the device to do?

A transistor can be used as:

- a controlled switch for digital logic;
- a small-signal amplifier;
- a power-control device;
- an analog current source or active load;
- part of an oscillator, regulator, sensor interface, or radio circuit.

There is no universally "best transistor." Device choice follows voltage, current, frequency, gain, input-drive, density, power, manufacturing, and cost requirements. A CPU asks for billions of tiny switches that can be fabricated together, switch quickly, and consume little static power. That requirement strongly favors CMOS.

BJT — bipolar junction transistor

An NPN or PNP BJT uses both electron and hole transport—hence *bipolar*. A small change in base-emitter voltage causes an exponential change in collector current. Introductory circuit analysis often describes a BJT as "current controlled" because useful collector current normally requires base current:

```
base drive → collector-emitter conduction
```

That phrase is a circuit-model convenience, not the complete device physics. BJTs provide useful gain and transconductance and remain important in analog, RF, high-speed, and mixed-signal circuits. Bipolar logic families also existed, including TTL and ECL. Their drive and static-power trade-offs are less suited to packing modern general-purpose CPUs at CMOS-scale density.

JFET — junction field-effect transistor

A JFET controls a semiconductor channel by expanding or shrinking a reverse-biased junction depletion region. Its gate normally draws very little steady current. A common JFET is naturally conducting at zero gate-source voltage, so an applied gate voltage pinches the channel toward cutoff:

```
zero gate bias → channel normally available
reverse gate bias → depletion region grows → channel narrows
```

JFETs remain useful where low noise, high input impedance, or particular analog behavior matters. They are not the dominant device for dense complementary digital logic.

MOSFET — metal-oxide-semiconductor field-effect transistor

The MOSFET controls a channel through an insulated gate. Changing gate voltage moves charge and establishes an electric field; ideally, maintaining a steady gate voltage needs almost no DC gate current.

Enhancement and depletion modes

- An **enhancement-mode** MOSFET is normally off at zero gate-source bias and requires the appropriate gate voltage to create or strengthen a channel. Modern CMOS logic primarily uses complementary enhancement-mode NMOS and PMOS devices.
- A **depletion-mode** MOSFET has a channel at zero gate-source bias and needs opposite-polarity control to reduce conduction. It appears in specialized circuits and historical logic styles but is not the default CMOS switch in this course.

NMOS and PMOS together

CMOS pairs complementary devices:

```
input low:   PMOS pull-up conducts, NMOS pull-down stops
input high:  NMOS pull-down conducts, PMOS pull-up stops
```

Ideally, a stable inverter output has no direct conducting path from supply to ground. Real CMOS still consumes leakage power and consumes dynamic energy while capacitances charge and discharge. CMOS won because complementary restoring logic combines manufacturability, density, noise margin, and relatively low static power.

Planar MOSFET, FinFET, and GAAFET

These names describe evolving MOSFET geometries, not three unrelated Boolean devices.

Planar MOSFET

In a planar device, the channel lies near the flat silicon surface and the gate controls it mainly from one side. As dimensions shrank, electrostatic control became harder: leakage and short-channel effects increased.

FinFET

A FinFET raises the channel into a narrow fin. The gate wraps around multiple sides of that fin:

```
planar: gate over channel
FinFET: gate around the fin's sides and top
```

The stronger electrostatic control reduces leakage and supports continued scaling. Drive strength can be adjusted partly through the number and geometry of fins. To software, it is still hidden beneath the same logic-gate and ISA abstractions.

GAAFET / nanosheet transistor

A gate-all-around FET surrounds the channel more completely. Modern forms may use stacked horizontal nanosheets or nanowires. Greater gate coverage improves electrostatic control as dimensions continue shrinking.

FinFET and GAAFET technology changes delay, leakage, voltage, density, variability, and manufacturing constraints. It does not change the meaning of an `ADD` instruction. Those effects influence the physical implementation that must satisfy the architecture.

Power-device relatives

Power MOSFETs and IGBTs are optimized for switching substantial voltage and current in converters, motor drives, and power systems. An IGBT combines a MOS-gated input with bipolar conduction characteristics. These devices solve a different problem from dense on-chip CPU logic: safely controlling energy rather than integrating billions of minimum-size logic switches.

The useful systems-programmer mental model

```
device family and geometry
  → voltage/current/leakage/delay characteristics
  → logic-gate and storage-cell implementation
  → datapath, cache, clock, and power constraints
  → microarchitecture
  → ISA-visible behavior
  → compiler, kernel, and applications
```

Kernel code normally observes the upper layers: architectural registers, memory ordering, timers, performance counters, thermal/frequency behavior, machine checks, and device interfaces. The transistor family influences those constraints, but software does not command an individual FinFET or GAAFET.

Keep these distinctions

- **BJT, JFET, and MOSFET** identify different device structures and control behavior.
- **NMOS and PMOS** are complementary MOSFET polarities used together in CMOS.
- **Enhancement and depletion** describe whether a MOS channel is normally absent or present under the reference bias.
- **FinFET and GAAFET** are improved MOSFET geometries for electrostatic control and scaling.
- **CMOS** is a complementary circuit-design approach, not one transistor.
- None of these names alone determines CPU performance; circuit, microarchitecture, process, packaging, memory, workload, cooling, and software all matter.

Selected references

- Patt and Patel, *Introduction to Computing Systems*, device-to-logic sections for the digital abstraction.

- R. Jacob Baker, *CMOS: Circuit Design, Layout, and Simulation*, selected MOSFET and CMOS chapters.
- Weste and Harris, *CMOS VLSI Design*, selected device, inverter, scaling, and circuit chapters.
- Current foundry or processor-vendor technology explanations when a specific process generation matters.